System in 2016, followed by the achievement of the highest level of safety certification from independent international assessors in 2019. By 2020, Kavach was approved as the national ATP system. Despite setbacks like the Covid-19 pandemic, the system pushed the project forward. Funding has been sanctioned, and there is significant political support as well.

SO, WHAT'S TAKING TIME?

Indian Railways stresses that installing such a complex safety system takes time. "It involves installation of towers and laying of optical fibre cable alongside the entire track length, installation of Loco Kavach on each locomotive and Station Kavach at each station," says a railways ministry spokesperson. "It is extremely complex, particularly when it is to be installed on existing network in train-running conditions." But that may not be the complete picture.

Railways may have designed the system and is the owner of the tech, but Kavach components are manufactured by private companies. Currently, there are only three companies supplying the components: Medha Servo Drives, Kernex Microsystems and HBL Power Systems. Four more firms have been vying for the approval to be vendors: G.G. Tronics, Quadrant Future Tek, Areca Embedded Systems and the state-owned BHEL. But sources say movement has been slow. Japan's Kyosan, which supplies signalling systems to the railways, and Germany's Siemens, are also working on Kavach, officials say.

After the recent accident, Railway Board CEO and chairperson Jaya Varma Sinha said that vendors need to ramp up their production capacity for the rollout to happen all over India. However, achieving this is easier said than done, considering the involvement of multiple agencies and technical intricacies. The railways' technical research arm, the Research Designs and Standards Organisation (RDSO), jointly holds the intellectual property with private players. As is the norm in railways, it is the RDSO that approves these products, their designs and efficacy. On top of that, the components need to have seamless interoperability, which means that components need to be able to talk to each other even if they are manufactured by different companies. That requires rigorous, on-ground testing. Then there is the Centre of Excellence for Kavach. Housed at the Indian Railways Institute of Signal Engineering & Telecommunication in Secunderabad, this centre exists to train and develop the technology for Kavach. It remains in a constant loop throughout the implementation process, gathering feedback, participating in field trials, and so forth. However, the centre is hobbled by the fact that it serves primarily in an advisory capacity.

Those in the know say, on condi-

KAVACH: THE DE

WHAT IS KAVACH?

An Automatic Train Protection system developed by the Research Design and Standards Organisation of the Ministry of Railways in collaboration with private companies

IT ENHANCES SAFETY BY...

- Preventing trains from jumping signals
- Averting collisions between trains on the same track
- Ensuring trains do not run on the wrong side
- Displaying signal information to loco pilots
- Sounding warning whistles at level crossings



KRISHNENDU HALDER

1,465 km

The length of railway route—over 2% of the total route length—on which Kavach has been implemented

₹44,640 cr.

The estimated cost of installing Kavach on India's total route length of 68,000 km and the entire fleet of 15,200-strong diesel and electric locomotives. (**Rs 34,000 cr.** for installation along the entire route length at Rs 50 lakh per route km and **Rs 10,640 cr.** for equipping all engines at Rs 70 lakh per unit)

WHY CAN'T IT BE SPEEDED UP?

- **Complex installation** Installation requires extensive infrastructure, like setting up towers and laying cables, while accommodating ongoing train operations
- **Limited manufacturing capacity:** Few approved manufacturers; components need interoperability testing, complicating production. Railways needs to onboard more players after testing
- **Bureaucratic layers** Involvement of multiple agencies—the RDSO, Centre of Excellence and Railway Board—in testing and approvals adds to delays
- **Technology maturity and resource constraints** Concerns about system reliability until as late as 2020 and significant logistical challenges hinder rapid rollout

Graphic by TANMOY CHAKRABORTY